

1.9 The Matrix of a Linear Transformation

The discussion that follows shows that every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ is actually a matrix transformation $\vec{x} \mapsto A\vec{x}$ and that important properties of T are intimately related to familiar properties of A . The key to finding A is to observe that T is completely determined by what it does to the columns of the $n \times n$ identity matrix I_n .

Example 1

The columns of $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ are $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Suppose T is a linear transformation from $\mathbb{R}^2$ into $\mathbb{R}^3$ such that

$$T(\vec{e}_1) = \begin{bmatrix} 5 \\ -7 \\ 2 \end{bmatrix} \text{ and } T(\vec{e}_2) = \begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix}$$

Without any additional information, find a formula for the image of an arbitrary $\vec{x}$ in $\mathbb{R}^2$.

Solution

Write

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = x_1 \vec{e}_1 + x_2 \vec{e}_2 \quad (1)$$

Since T is a linear transformation

$$\begin{aligned} T(\vec{x}) &= x_1 T(\vec{e}_1) + x_2 T(\vec{e}_2) \\ &= x_1 \begin{bmatrix} 5 \\ -7 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix} = \begin{bmatrix} 5x_1 - 3x_2 \\ -7x_1 + 8x_2 \\ 2x_1 + 0x_2 \end{bmatrix} \end{aligned} \quad (2)$$

The step from equation (1) to equation (2) explains why knowledge of $T(\vec{e}_1)$ and $T(\vec{e}_2)$ is sufficient to determine $T(\vec{x})$ for any $\vec{x}$. Moreover, since (2) expresses $T(\vec{x})$ as a linear combination of vectors, we can put these vectors into the columns of a matrix A and write (2) as

$$\begin{aligned} T(\vec{x}) &= \begin{bmatrix} T(\vec{e}_1) & T(\vec{e}_2) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= \begin{bmatrix} 5 & -3 \\ -7 & 8 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = A\vec{x} \end{aligned}$$

1.9 The Matrix of a linear transformation (cont'd)

Theorem 10

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that

$$T(\vec{x}) = A\vec{x} \quad \text{for all } \vec{x} \text{ in } \mathbb{R}^n$$

In fact, A is the $m \times n$ matrix whose j th column is the vector $T(\vec{e}_j)$, where $\vec{e}_j$ is the j th column of the identity matrix in $\mathbb{R}^n$.

$$A = [T(\vec{e}_1) \dots T(\vec{e}_n)] \quad (3)$$

Example 2

Find the standard matrix A for the dilation transformation $T(\vec{x}) = 3\vec{x}$, for $\vec{x}$ in $\mathbb{R}^2$.

Solution

So, any input vector is magnified three times. In $\mathbb{R}^2$, there are two basis vectors, $\vec{e}_1$ and $\vec{e}_2$.

We can write

$$A = [3T(\vec{e}_1) \quad 3T(\vec{e}_2)] = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

Example 3

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the transformation that rotates each point in $\mathbb{R}^2$ about the origin through an angle φ , with counterclockwise rotation for positive angles. We could show geometrically that such a transformation is linear. Find the standard matrix A of this transformation.

Solution

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ rotates into $\begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix}$, and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ rotates into $\begin{bmatrix} -\sin \varphi \\ \cos \varphi \end{bmatrix}$. See

Figure 1.

By theorem 10, $A = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix}$

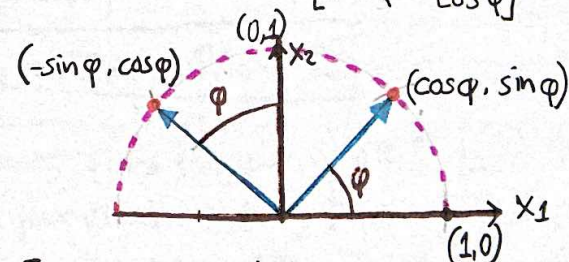


Figure 1 - A rotation transformation

1.9 The Matrix of a Linear Transformation (cont'd)

III

Existence and Uniqueness Questions

The concept of linear transformation provides a new way to understand the existence and uniqueness questions asked earlier. The next two definitions give the appropriate terminology for transformations.

Definition

A mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be onto $\mathbb{R}^m$ if each $\vec{b}$ in $\mathbb{R}^m$ is the image of at least one $\vec{x}$ in $\mathbb{R}^n$.

Definition

A mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be one-to-one if each $\vec{b}$ in $\mathbb{R}^m$ is the image of at most one $\vec{x}$ in $\mathbb{R}^n$.

Example 4

Let T be a linear transformation whose standard matrix is

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

Does T map $\mathbb{R}^4$ onto $\mathbb{R}^3$? Is T a one-to-one mapping?

Solution

Since A happens to be in echelon form, we can see at once that A has a pivot position in each row.

By theorem 4 in Section 1.4 (p.62), which states that

d. A has a pivot in every row.

and

a. For each $\vec{b}$ in $\mathbb{R}^m$, the equation $A\vec{x} = \vec{b}$ has a solution.

we can draw the conclusion that for each $\vec{b}$ in $\mathbb{R}^3$, the equation $A\vec{x} = \vec{b}$ is consistent (meaning it has solutions), either one or infinitely many). (p.28).

This means we can map any $\vec{x}$ in $\mathbb{R}^4$ onto a $\vec{b}$ in $\mathbb{R}^3$ by help of the standard matrix A .

So, yes, T maps $\mathbb{R}^4$ onto $\mathbb{R}^3$.

However, since the equation $A\vec{x} = \vec{b}$ has a free variable (there are four variables and only three basic variables), each $\vec{b}$ is the image of more than one $\vec{x}$. T is not one-to-one.

Let's see if we can show this.

Say we want to map a vector $\vec{x}_1$ onto $\vec{u} = \begin{bmatrix} -2 \\ 5 \\ 5 \end{bmatrix}$

Solve $A\vec{x} = \vec{u}$

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1.9 The Matrix of a Linear Transformation (cont'd)

IV

$$x_1 - 4x_2 + 8x_3 + x_4 = -2$$

$$2x_2 - 1x_3 + 3x_4 = 5$$

$$5x_4 = 5$$

$$\boxed{x_4 = 1}$$

$$2x_2 - x_3 + 3 = 5$$

$$2x_2 - x_3 = 2$$

$$2x_2 = 2 + x_3$$

$$\boxed{x_2 = \frac{x_3}{2} + 1}$$

$$x_1 - 4\left(\frac{x_3}{2} + 1\right) + 8x_3 + 1 = -2$$

$$x_1 - 2x_3 - 4 + 8x_3 + 1 = -2$$

$$\boxed{x_1 = 1 - 6x_3}$$

$$x_3 = \text{free variable} = t$$

$$\vec{x}_A = \begin{bmatrix} 1 - 6t \\ 1 + \frac{t}{2} \\ t \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} -6 \\ +1/2 \\ 1 \\ 0 \end{bmatrix}$$

$$A\vec{x}_A = \begin{bmatrix} -2 \\ 5 \\ 5 \end{bmatrix}$$

Now, let's find another vector in $\mathbb{R}^4$ that equals $A\vec{x}_A$. suppose we start with $t=0$:

$$\vec{x}_A = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \text{ and } t=13 \Rightarrow \vec{x}_A = \begin{bmatrix} -77 \\ 15/2 \\ 13 \\ 0 \end{bmatrix}$$

$$\text{Both will map to } \begin{bmatrix} -2 \\ 5 \\ 5 \end{bmatrix}$$

Theorem 11

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then T is one-to-one if and only if the equation $T(\vec{x}) = \vec{0}$ has only the trivial solution (if $\vec{x} = \vec{0}$, also called the zero solution). p. 70.

Remark: To prove a theorem that says "P is true if and only if Q is true", one must establish two things:

- (1) If P is true then Q is true, and
- (2) If Q is true then P is true

Theorem 12

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, and let A be the standard matrix for T . Then:

- a. T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if the columns of A span $\mathbb{R}^m$.
- b. T is one-to-one if and only if the columns of A are linearly independent.

1.9 The Matrix of a Linear Transformation (cont'd)

Example 5

Let $T(x_1, x_2) = (3x_1 + x_2, 5x_1 + 7x_2, x_1 + 3x_2)$. Show that T is a one-to-one transformation. Does T map $\mathbb{R}^2$ onto $\mathbb{R}^3$?

Solution

The vectors in T are written in (x_1, x_2) notation. Let's write them as column vectors in a matrix

$$T(\vec{x}) = \begin{bmatrix} 3x_1 + x_2 \\ 5x_1 + 7x_2 \\ x_1 + 3x_2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 5 & 7 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (4)$$

By Theorem 12b, T is one-to-one. Theorem 12b states that

" T is one-to-one if and only if the columns of A are linearly independent". Which they are:

Does it map $\mathbb{R}^2$ onto $\mathbb{R}^3$? Theorem 4a can be used

here; the columns of A span $\mathbb{R}^3$ if and only if A has 3 pivot positions. This is impossible since A has 2 columns.

So no, T does not map $\mathbb{R}^2$ onto $\mathbb{R}^3$.

The columns of A spans a plane in $\mathbb{R}^3$.